

Manaswini Gande

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Professional Experience

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- BMO Bank | Coolsnail Technologies | AZURE Data Engineer | Chicago, Illinois, USA | Sep 2024 – Present**
- Developed and optimized scalable **ADF pipelines** and **ETL** workflows to ingest, transform, and load financial data into ADLS and Azure Synapse Analytics for real-time banking operations.
 - Implemented performance tuning using ADF, **Azure Databricks**, and **PySpark** to ensure low-latency, high-throughput data pipelines supporting mission-critical workloads.
 - Designed and managed data models and data warehouses using **Azure Synapse**, SQL Server, and Azure **SQL** Database to support key business intelligence and analytics requirements for financial reporting and regulatory compliance.
 - Established and managed Azure Storage Accounts, Blob Storage, and Data Lake **Gen2**, ensuring secure and scalable data storage solutions for sensitive financial transactions and customer data and worked closely with data science teams to operationalize machine learning models using **Azure Machine Learning**, Databricks **MLflow**, and Azure Synapse Analytics, embedding predictive models into bank data pipelines for fraud detection and risk assessment.
 - Streamlined CI/CD pipelines for data engineering workflows using **Azure DevOps**, Git enabling continuous integration and seamless deployment of data solutions in a secure, version-controlled environment.
 - Automated data extraction, cleaning, and orchestration in banking pipelines using **Python**, **SQL**, Scala, Pandas, and **NumPy**, delivering high-quality datasets for analytics and reporting.
 - Leveraged Power BI for building interactive dashboards and financial reports, utilizing data from Azure Synapse
 - Collaborated with engineering teams to implement Azure **Kubernetes** Service (**AKS**) and Docker for containerized solutions, enabling scalable deployments.
- Regeneron Pharmaceuticals | AWS Data Engineer | Tarrytown, New York, USA | Jan 2024 – Aug 2024**
- Utilized Sqoop and Impala to facilitate data migration between relational databases and Hadoop ecosystems using EMR.
 - Developed and managed ETL pipelines using Informatica and Python-based frameworks to extract, transform, and load data from multiple pharmaceutical systems into Snowflake and **Amazon S3**, ensuring data integrity.
 - Wrote optimized SQL queries and used Python for data extraction, transformation, and processing from relational databases like **MySQL**, **PostgreSQL**, and **NoSQL** databases like MongoDB to support advanced pharmaceutical analytics.
 - Utilized **PySpark** in AWS EMR to perform large-scale data transformations, enabling analytics and insights from pharmaceutical datasets for clinical trials, drug research, and regulatory reporting.
 - Designed and implemented real-time data streaming solutions using Kafka and Flume to integrate structured and unstructured pharmaceutical data into Snowflake and Amazon **RDS**.
 - Ensured data security, integrity, and compliance with regulatory standards like **HIPAA** by implementing encryption and access control mechanisms in **AWS Glue** and **Amazon S3**. Maintained audit logs and secured sensitive healthcare data.
 - Automated and orchestrated ETL workflows using Apache Airflow and AWS Glue, ensuring seamless scheduling and execution of data processing pipelines in cloud environments.
 - Managed distributed data storage using **HBase** and Cassandra on AWS EMR, coordinated using Zookeeper to ensure high availability and fault tolerance.
 - Collaborated with data scientists and analysts to deliver predictive insights using **AWS EMR** and PySpark. Built and deployed models for drug efficacy studies, patient risk analysis, and healthcare outcome predictions.
 - Monitored, troubleshooted, and optimized **ETL** performance using Amazon Redshift, SQL tuning, and EMR profiling to ensure cost-efficient data processing and high-performance analytics.
- Infosys | GCP Data Engineer | Hyderabad, India | Feb 2022 – Aug 2023**
- Used **Sqoop** on **Dataproc** to migrate legacy on-prem relational data into **Google Cloud Storage** and processed it further using Hive for integration with downstream analytics pipelines.
 - Developed and optimized ETL and **ELT pipelines** using **Cloud Dataflow** (Apache Beam) to process and validate structured and semi-structured data from **GCS** into **BigQuery**.
 - Designed Hive and Spark-based workflows using Cloud Dataproc to transform large-scale datasets and support advanced analytics use cases across marketing and operations.
 - Wrote Python scripts for preprocessing, validation, and transformation of data files ingested into BigQuery using Dataflow and scheduled via Composer.

- Deployed and managed batch and streaming workflows using Apache Airflow in Cloud Composer, leveraging custom Python operators and **BigQuery/Dataproc** hooks.
- Used Cloud Shell for provisioning services, running deployments, and automating operational tasks.
- Implemented row-level security policies in **Tableau** connected to **BigQuery** to ensure secure data access and compliance.
- Built dashboards and reports in **Looker Studio** (Data Studio) for monitoring project billing, service utilization, and cost optimization strategies.
- Collaborated with data scientists and analysts to provision datasets and build feature engineering pipelines for ML models using BigQuery and Dataproc.

Thales Group | Bangalore, India | Data Engineer INTERNSHIP

Mar 2020 – Jan 2022

- Developed batch data transformation jobs using PySpark on AWS **EMR** to clean, enrich, and join large pharmaceutical datasets from multiple systems.
- Utilized Spark SQL and DataFrames to implement business rules, perform aggregations, and calculate metrics required for reporting and downstream analytics.
- Processed semi-structured and unstructured data stored in **HDFS** and **Amazon S3**, transforming them into structured formats using custom Python scripts.
- Executed table-to-table transformation workflows within **Spark** to support **CDC** (Change Data Capture) pipelines and incremental data loads.
- Integrated **Kafka** with **Spark** jobs to process real-time and near-real-time message streams for ingesting updates from operational systems.
- Leveraged MapReduce on **Hadoop** to process image and binary files for classification, **metadata** tagging, and storage in optimized partitions.
- Designed data parsing logic to normalize raw logs, nested **JSON**, and **CSV** inputs into consistent, query-ready tables stored in S3 and Hive.
- Performed format conversion tasks including Parquet, **ORC**, and **Avro** for compression optimization and query efficiency in downstream tools.

TECHNICAL SKILLS

Big Data Ecosystem: HDFS, Yarn, MapReduce, Spark, Kafka, Kafka Connect, Hive, Airflow, Stream Sets, Sqoop, HBase, Flume, Pig, Ambari, Oozie, Zookeeper, Nifi, Sentry

Hadoop Distributions: Apache Hadoop 2.x/1.x, Cloudera CDP, Hortonworks HDP

Cloud Environment: Amazon Web Services (AWS), Microsoft Azure, GCP

Databases: MySQL, Oracle, Teradata, MS SQL SERVER, , DB2, Mongo DB

AWS: EC2, EMR, S3, Redshift, EMR, Lambda, Kinesis Glue, Data Pipeline

Microsoft Azure: Databricks, Data Lake, Blob Storage, SQL Database, SQL Data Ware- house, Cosmos DB,

Operating systems: Linux, Unix, Windows 10, Windows 8, Windows 7, Windows Server 2008/2003, Mac OS

Reporting Tools/ETL Tools: Informatica, Talend, SSIS, SSRS, SSAS, ER Studio, Tableau, Power BI, Arcadia, Data stage,

Programming Languages: Python (Pandas, SciPy, NumPy, Scikit-Learn, Stats Models, Matplotlib, Plotly, TensorFlow, PyTorch), PySpark, T-SQL/SQL, PL/SQL, HiveQL, Scala, UNIX Shell Scripting, C#, Java, Angular.

Version Control: Git, SVN, Bitbucket

Development Tools: Eclipse, NetBeans, IntelliJ, Hue, Microsoft Office

EDUCATION

- **Masters in computer science** from Northern Arizona University – Arizona USA(Aug 2023-May 2025)
- **Bachelors in computer science and engineering** – Telangana, India (Aug 2017- Sept 2021)

CERTIFICATIONS

- Microsoft Certified: Azure Data Fundamentals
- Google Cloud Certified: Associate Cloud Engineer